2026 Geophysical Characterization of the Historic Onyx Mine in Steamboat Springs, CO

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Abstract

Non-invasive geophysical techniques can be used to identify and characterize subsurface structures such as mine shafts, which pose potential public safety hazards if left undocumented. This study used ground-penetrating radar (GPR) and gravity surveys to investigate a known mine shaft at the historic Onyx Mine near Steamboat Springs, Colorado. Results suggest the presence of subsurface variations near the known mine shaft location; however, limitations in survey design and the conductive, clay-rich lithology reduced GPR penetration and prevented definitive shaft identification. Because geophysical interpretations are non-unique, the mine shaft cannot be conclusively characterized using the collected data alone. Future investigations should integrate electrical resistivity tomography (ERT), GPR, and gravity methods to improve the detection and characterization of the mine shaft structures.

Introduction

It is an important and difficult feat to locate subsurface mine shafts. Many historic mine shafts are unknown or unlocated in modern day. Over time, mine shafts become unstable and can collapse. The collapse of a mine shaft can develop into a sinkhole, which poses a severe public safety hazard. The mine shaft atop Howelsen Hill, indicated by the yellow star in Fig. 1, is located beneath extensive bike trail networks. Therefore, it is paramount that we gain an understanding of these mine shaft structures to ultimately determine the safety risk these mine shafts pose to the people of Steamboat Springs, CO.

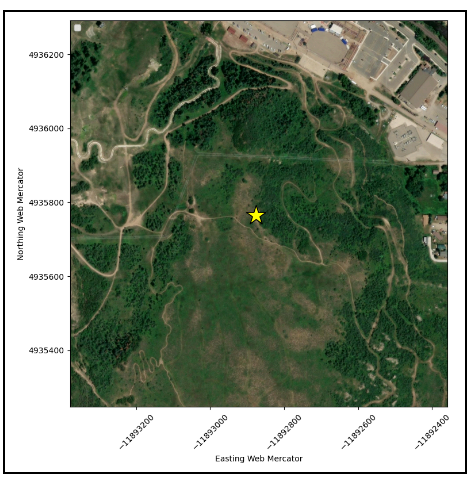

Figure 1: Satellite view of survey site.

Howelsen Hill is located atop the Browns Park geological formation, meaning we expect to see sandstone, limestone, and some conglomerates in the subsurface. From historical papers and articles, we were able to determine our site's specific near-surface lithology. In a paper from 1898 by the Iowa Academy of Science, the mine shaft walls were described as being packed in with clay [5]. Another paper from *Steamboat Pilot and Today* confirms the presence of clay on Howelson Hill, more specifically, a 2-3 foot layer of unconsolidated soil over a layer of clay in the shallow-subsurface [4]. We deduced that GPR and Gravity would be the most effective methods for locating and characterizing this subsurface mine shaft.

Methods

Local variations in subsurface material properties affect the way signals propagate through the ground. When a void is present in the subsurface, the boundary between the air-filled void and solid rock has a large contrast in material properties that geophysical methods can detect.

A. Gravity

Gravity is controlled by the distribution of mass in the subsurface. Gravimeters measure variations in the Earth's gravitational acceleration due to the underlying and surrounding mass bodies. In equation 1, we observe that the gravitational acceleration (g) is controlled by the gravitational constant (G), the mass causing the gravity field (M), and the distance between the gravimeter and that mass (r). This implies that more mass results in higher gravitational acceleration values, and less mass results in lower gravitational acceleration values.

$$g = \frac{GM}{r^2} \tag{1}$$

The gravity survey included a 29 m line (going SE-NW) with each measurement spaced 1m across the entire line. We incorporated denser spacing around the target, from 12 m to 16 m, 0.5 m spacing was used. Two separate CG5's were used to take these measurements, each starting on either side of the survey line.

Once we obtained the raw gravity data, we did linear drift, instrument-tie, free-air, Bouguer, and LiDAR terrain corrections. The linear drift correction was done by visiting a base station at the beginning and end of the survey to account for any gravitational drift through time. The instrument-tie correction was done by having an overlapping station measured by each CG5 gravimeter used. Free-air and Bouguer corrections were implemented to account for the gravimeter's distance from the center of the Earth and the mass density throughout that distance. Finally, LiDAR terrain corrections were done to account for the elevation and topographic contrasts around the survey site.

We noticed that our gravity profile, once these corrections were done, was still being controlled by the hillside/cliff next to our survey line. The presence of the cliff was giving us terrain-induced negative anomalies that we, initially, incorrectly interpreted as the mine shaft anomaly.

We went back in and checked our GPS station elevations that we used to do the free-air and Bouguer corrections, and found a meter jump between two stations, which was unrealistic. We interpolated the correct elevations from the LiDAR DEM and got a smooth, accurate station elevation profile. Fig. 2 shows the initial elevation profile, compared to the corrected elevation.

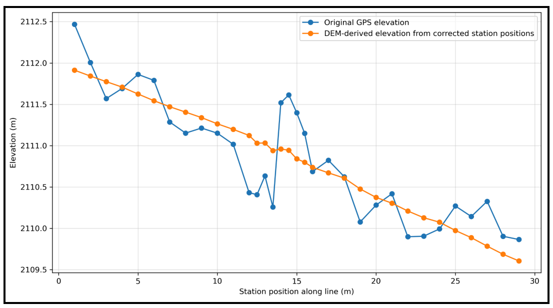

Figure 2: Original elevations compared to DEM-corrected elevations

Once this elevation correction was completed, we went back and did accurate free-air, Bouguer, and LiDAR terrain corrections to produce our final residual gravity profile.

B. GPR

Ground Penetrating Radar (GPR) is sensitive to the relative dielectric permittivities (ϵ_r) and magnetic permeabilities (μ) of the materials the signals propagate through, and these two variables make up the impedance (Z) of a material, see equation 2. When the electromagnetic (EM) signal sent out by the transmitter of the GPR reaches a boundary between two materials that have differing properties, there is a resulting impedance contrast at this boundary. The magnitude of the impedance contrast dictates the strength of the reflector captured by the GPR receiver.

$$Z = \sqrt{\frac{\mu}{\epsilon_r}} \qquad (2)$$

The GPR survey included 2 lines of approximately 60 m, one line heading in the SE direction and the other tracing over the first heading in the NW direction. This survey included two Impulse Radars, each with two frequency antennas. The low-frequency impulse radar had 70 MHz and 300 MHz antennas at a 5 cm trigger distance. The higher-frequency impulse radar had 170 MHz and 600MHz antennas at a 2 cm trigger distance.

After data collection, we processed the raw radargrams through the software program Geolitix. The following corrections were applied to our radargrams: time cut, time zero correction, manual gain, background subtraction, frequency filter, and a constant scale. This provided clearer subsurface reflections in the processed radargram, which allows us to interpret what is happening in the subsurface.

Fig. 3 shows our survey design for both gravity and GPR. The mine shaft opening and tunnel are perpendicular to the gravity and GPR survey lines depicted by blue dots and red/purple lines, respectively. These survey lines are overlaid on a topographic map, showing the hillside that could be affecting the data results.

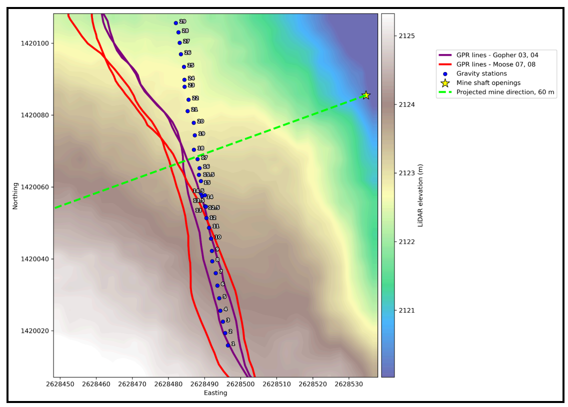

Figure 3: Map of GPR & Gravity surveys above the historic Onyx Mine atop Howelsen Hill, Steamboat Springs, CO.

Results & Discussion

A. Gravity

Fig. 4 shows the final residual gravity profile after processing. The final gravity results are inconclusive as to whether we identified a structured mine shaft, a collapsed mine shaft, or the correct mine shaft location. We found that the terrain was largely influencing our gravity anomalies. After extensive terrain corrections were applied, any negative anomalies previously interpreted as a mine shaft were no longer present in the gravity profile. This led us to believe that the single gravity survey line does not provide enough data to uniquely claim anything about the mine shaft.

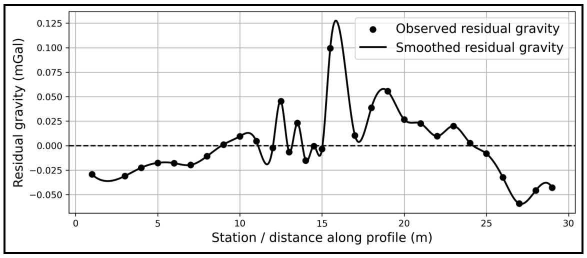

Figure 4: Final residual gravity anomaly.

In an ideal case, we would expect to see a clear negative residual gravity anomaly, with this anomaly correlating to the mine shaft/void space location. It is clear that the processed gravity profile, in Figure 4, does not show any indication of a negative gravity anomaly.

B. GPR

Our final processed radargrams from GPR do not identify where mine shafts could be present. From background geology, as well as a clear attenuating signal in our GPR profile around 50 ns (shown in Fig. 5), we predict a highly conductive clay layer to be present here. This would explain our loss of signal in the near-subsurface area. Due to the clay layer attenuating most of our signals in the first 2.5-3 meters of our radargrams, we are not able to map the predicted 4-10 meter ceiling of the mine shaft.

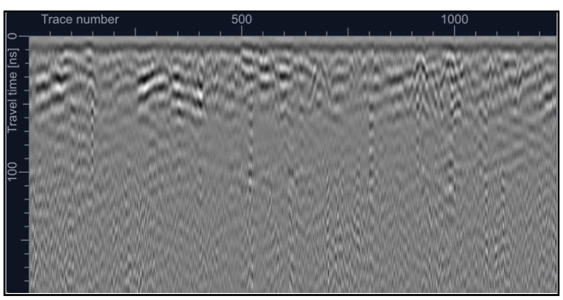

Figure 5: GPR radargram on line 007, using a 70 MHz frequency, after being processed.

Although our results are inconclusive on all accounts, we must consider the survey parameters that were carried out in the field to produce these results. We can utilize this knowledge to design a survey that can be done in the future with higher resolutions, more mine shaft coverage, and a higher likelihood of identifying the onyx mine. The goal is to classify the historic mine's current structural state to determine whether it poses a safety risk.

Future Work

To improve upon this survey in order to get conclusive results, we recommend that future teams revisit the onyx mine and conduct a 3 method survey utilizing gravity, GPR, and ERT. The recommended survey plans are shown in Fig. 6.

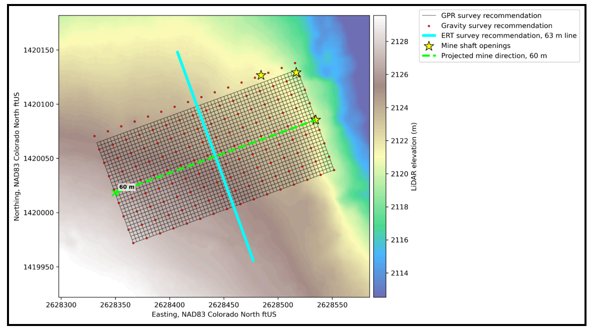

Figure 6: Map of recommended GPR, Gravity, and ERT surveys above the historic Onyx Mine atop Howelsen Hill, Steamboat Springs, CO

A. Gravity

Teams should construct a 60 m x 30 m grid with lines extending NW-SE perpendicular to lines extending NE-SW atop the Onyx Mine. This survey uses 4m spacing between gravimeter stations along these lines. We have found evidence for successful mine shaft detections using a grid with 5 m spacing, which supports our chosen 4m spacing [1]. GPS measurements at every gravimeter station are required for accurate terrain corrections. There must also be visits to the base station at the beginning and end of the survey to account for linear drift. If the survey takes multiple days, then return to the base station every few hours and at the start and end of each field day. Teams should utilize both of the CG-5's available to them for efficiency. If all of these survey parameters are followed, the survey is estimated to take 5-6 hours.

B. GPR

An updated GPR survey includes constructing a 60 m x 30 m grid atop the same grid as the gravity survey in figure 6. GPR should use 0.5 m line spacing across this grid. It was determined that a GPR with a frequency of 25 MHz would be the most effective at this specific site. This frequency has the potential to reach depths of mine shafts and have less attenuation in the highly conductive clay layer. The survey is estimated to take 1-2 hours.

C. ERT

In the center of the gravity/GPR grid, set up a 63 m line using 64 nodes at 1 m spacing and run a Dipole-Dipole array with an ABEM WalkTEM-2. We recommend a

Dipole-Dipole array because it is the fastest and has the highest lateral resolution compared to other arrays. A secondary array to run, if time allows, would be the Wenner array. This array is effective because it has a large signal-to-noise ratio and a strong ability to resolve vertical soil stratification. Each array is expected to take 1-2 hours, along with the time it takes to set up the survey.

Conclusion

Overall, gravity and GPR show potential for characterizing the historic Onyx mine, but the data collected in this survey are not sufficient to definitively identify or characterize the mine shaft. Because both methods produced non-unique results and had limited survey coverage, we cannot determine from the current data alone whether the shaft is open, collapsed, or offset from the projected extent into the hillside. Future work should include the use of a gravity grid, lower-frequency GPR, and ERT to improve subsurface coverage and better constrain the mine's geometry. This integrated approach would provide a stronger basis for evaluating whether the Onyx mine poses a current safety risk.

Author Statement

Gravity measurements were acquired by groups 2 and 4 in the 2026 Mines Geophysics Field Camp, and GPR measurements were acquired by group 2, including M.B. and T.M. Gravity processing and corrections were mainly done by T.M, with a few contributions from S.L. Survey maps and parameters were completed by M.B. GPR processing was completed by S.L. The interpretation of our results was done by M.B, S.L, and T.M, all equally contributing. T.M. created and set up the project's GitHub on June 2, 2026[13], while M.B, S.L, and T.M. contributed extensively throughout the project.

Acknowledgments

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